Fluid Mechanics Lab Project Report

***Title**: Comparative Study of Flow Measurement Using Triangular Notch and Trapezoidal Notch in Open Channel Flow*

Class: BEME-F-24-B

Group Members:

Sr.No	Names	Roll No.s
1	Muhammad Usman Khan	241202
2	Touseed Ahmed Khan	241196
3	Alishba Saeed	241212
4	Areeba	241253

Submitted To: Sir Aamir Qamir

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PROJECT TITLE

Comparative Study of Flow Measurement Using Triangular Notch and Trapezoidal Notch in Open Channel Flow

Abstract:

Flow measurement is one of the most important applications of fluid mechanics in hydraulic and industrial systems. Different types of notches are commonly used to measure discharge in open channels depending on the flow conditions and required accuracy. In this project, the performance of a triangular notch and a trapezoidal notch was studied experimentally as well as theoretically. The main objective of this project was to understand how notch geometry affects discharge measurement and to compare the accuracy, sensitivity, and practical applications of both notch types. Experimental readings were taken for different values of head, and the corresponding discharge values were calculated. The obtained results were then compared with theoretical predictions. The project helped in understanding the practical implementation of open channel flow measurement and the effect of notch design on hydraulic performance.

Introduction:

In fluid mechanics, accurate measurement of flow rate is necessary for the design, operation, and control of hydraulic systems. In open channel flow, one of the simplest and most reliable methods of measuring discharge is by using notches. A notch is basically an opening of a specific shape provided in a vertical plate through which water flows.

Different notch shapes are used depending on the range of flow and required accuracy. Among these, the **Triangular Notch** is widely used for measuring small flow rates because it is highly sensitive to small changes in head. On the other hand, the **Trapezoidal Notch** is commonly used for medium and large discharge measurements because it combines the advantages of both rectangular and triangular notches.

This project focuses on studying the working behavior of both notch types under different flow conditions and comparing their hydraulic performance.

Aim of Project:

The main aim of this project is to study, analyze, and experimentally compare the performance of triangular and trapezoidal notches for open channel flow measurement.

Objectives:

Primary Objectives

- To study the working principle of notches.
- To understand the relationship between head and discharge.
- To compare the hydraulic performance of triangular and trapezoidal notches.

Secondary Objectives

- To perform theoretical discharge calculations.
- To collect experimental discharge data.
- To evaluate measurement accuracy.
- To identify practical engineering applications.

Theoretical foundation

In fluid mechanics, flow measurement is an important process used to determine the quantity of liquid passing through a channel per unit time. In many hydraulic systems such as irrigation canals, drainage systems, treatment plants, and laboratory setups, it is necessary to measure discharge accurately in order to ensure proper system operation and performance.

When water flows in an open channel, the liquid surface remains exposed to atmospheric pressure. Unlike closed pipe flow, the motion of fluid in open channels is mainly affected by gravity. For such applications, notches are commonly used because they provide a simple, economical, and reliable method of discharge measurement.

A notch is a sharp-edged opening made in a thin vertical plate through which liquid flows freely. By measuring the height of water above the notch crest, called head, the discharge can be determined using fluid mechanics relationships.

Basic Terminology of Notches

Before studying notch performance, it is important to understand the basic terms used in notch analysis.

- **Crest:** The lower horizontal edge of the notch from where water starts flowing.
- **Head:** The vertical distance between the free water surface and the crest of the notch. Head is the most important parameter because discharge directly depends on it.
- **Coefficient of Discharge:** The ratio of actual discharge to theoretical discharge. It accounts for practical losses such as friction, viscosity, contraction, and turbulence.
- **Nappe:** The sheet of water flowing over the notch is called the nappe.
- **End Contraction:** Reduction in effective flow width due to contraction of fluid near notch boundaries.

Working Principle of Notches

The operation of notches is based on the conversion of potential energy of water into kinetic energy. When water approaches the notch, it possesses potential energy due to its elevation above the crest. As water flows through the notch opening, this potential energy is converted into velocity.

The discharge through the notch depends on:

- Height of water above the notch
- Shape of notch opening
- Fluid properties
- Gravitational acceleration
- Flow conditions

By measuring the head and applying hydraulic equations, discharge can be determined accurately.

Classification of Notches

Common notch types include:

- **Rectangular Notch:** Used for medium discharge measurements.
- **Triangular Notch:** Used for low discharge measurements.
- **Trapezoidal Notch:** Used for moderate and high discharge measurements.
- **Circular Notch:** Used for specialized hydraulic applications.

Fluid Mechanics Principles Used

- **Bernoulli's Principle:** This principle relates pressure energy, kinetic energy, and potential energy in flowing fluids.
- **Continuity Equation:** Mass flow remains constant throughout the flow system.
- **Torricelli's Law:** The velocity of fluid through an opening depends upon the liquid head.

Literature review:

Previous Research on Notches

Previous hydraulic studies show that notch geometry significantly affects discharge accuracy.

Researchers have found that:

- The **Triangular Notch** provides higher sensitivity for small flow rates.
- The **Trapezoidal Notch** reduces end contraction effects.
- Experimental coefficients vary depending upon fabrication accuracy and flow conditions.

Comparison from Literature

Triangular Notch	Trapezoidal Notch
Advantages	
High sensitivity	Wide flow range
Better low-flow accuracy	Reduced contraction losses
Simple calibration	Better industrial suitability
Disadvantages	
Limited flow range	More complex geometry
Sensitive to installation errors	Slightly lower sensitivity at small heads

Equipment Required:

In order to complete the exercise, we need a number of pieces of equipment.

- Hydraulics Bench which allows us to measure flow by timed volume collection.
- Stilling baffle
- Trapezoidal and Vee Notches
- Vernier Height Gauge
- Stop Watch
- Spirit Level

Apparatus Overview:

The apparatus has five basic elements used in conjunction with the flow channel in the molded bench top of Hydraulics Bench Description.

- A stilling baffle and the inlet nozzle combine to promote smooth flow conditions in the channel
- A Vernier hook and point gauge is mounted on an instrument carrier, to allow measurement of the depth of flow above the base of the notch
- Finally, the weir notches are mounted in a carrier at the outlet end of the flow channel

To connect the delivery nozzle, the quick release connector is unscrewed from the bed of the channel and the nozzle screwed in place. The stilling baffle is slid into slots in the wall of the channel. These slots are polarized to ensure correct orientation of the baffle.

The instrument carrier is located on the side channels of the molded top. The carrier may be moved along the channel to the required measurement position. The gauge is provided with a coarse adjustment locking screw and a fine adjustment nut.

The Vernier is locked to the mast by screw and is used in conjunction with the scale. The hook and point are clamped to the base of the mast by means of a thumb screw.

The weir may be clamped to the weir carrier by thumb nuts the weir plates incorporate captive studs to aid assembly.

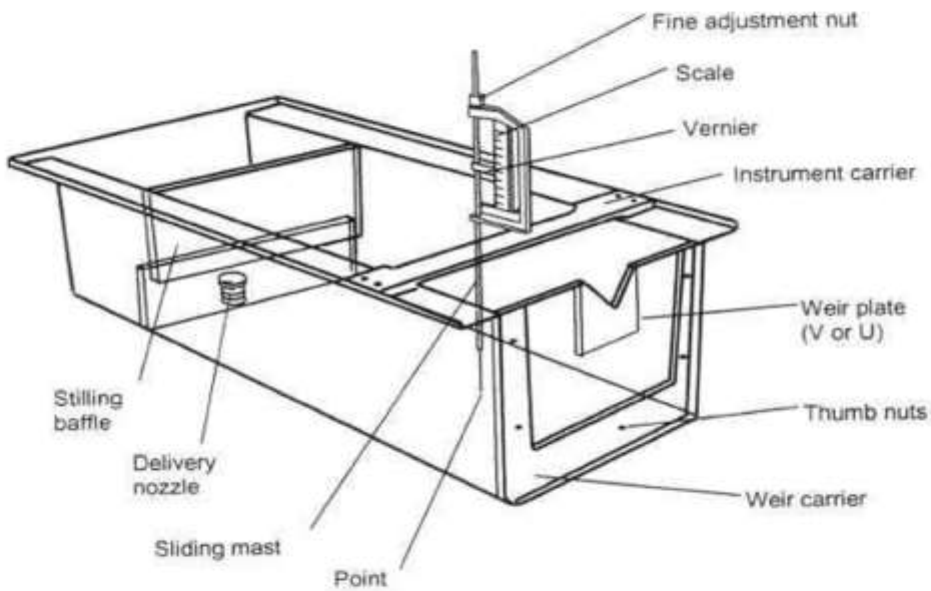


Figure 1

Experimental Setup:

Ensure that the hydraulic bench is positioned so that its surface is horizontal (necessary because flow over notch is driven by gravity). Mount the rectangular notch plate into the flow channel and position the stilling baffle as shown in the diagram. In order to measure the datum height (with the height gauge) of the base of the notch, position the instrument carrier the opposite way round from that shown in the diagram. Then carefully lower the gauge until the point is just above the notch base and lock the coarse adjustment screw. Then, using the fine adjustment, adjust the gauge until the point just touches the notch bottom and take a reading; be careful not to damage the notch.

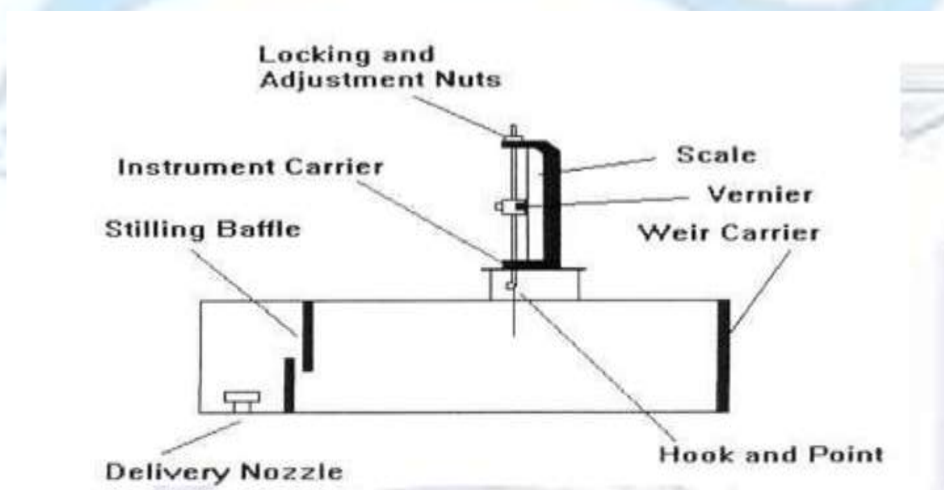


Figure 2

Mount the instrument carrier as shown in the diagram and locate it approximately half way between the stilling baffle and the notch plate. Open the bench control valve and admit water to the channel; adjust the valve to give approximately 10mm depth above the notch base. To help achieve this, you may find it useful to pre-set the height gauge position to give a rough guide.

Design and fabrication of notch plates

Design and fabrication of notch plates

The notch plates were designed based on standard hydraulic weir geometries and fabricated from steel sheets of **4 mm thickness**. The material density was taken as the average density of structural steel (**7850 kg/m³**), which was later used for weight estimation and handling considerations.

Design Specifications

- **Triangular (Vee) Notch:**
  - ✓ Notch angle (φ): **60°**
  - ✓ Plate outer dimensions: **0.24m x 0.07m**
  - ✓ Crest machined to a sharp edge at the vertex.
- **Trapezoidal Notch:**
  - ✓ Bottom crest width (B): **0.24m**
  - ✓ Side slope angle (φ): **30°**
  - ✓ Plate outer dimensions: **0.24m x 0.07m**

Fabrication Process

- **CNC Laser Cutting:** Both notch profiles were cut from the 4 mm steel plate using a CNC laser machine. This ensured precise initial dimensions, smooth cut edges, and minimal thermal distortion compared to manual cutting.
- **Drilling Operation:** After laser cutting, a drilling machine was used to create mounting holes on the upper sides of each plate. These holes allowed the notches to be securely clamped to the weir carrier in the experimental channel.
- **Dimensional Errors & Correction (Hand Grinding):** Upon measuring the cut notches, minor length deviations were observed due to laser kerf variations and slight thermal expansion. Both notches showed an error in the crest length and side edge dimensions. To correct this:
  - ✓ A **hand grinding machine** was carefully used to remove excess material and reset the critical lengths.
  - ✓ The grinding was performed gradually, with frequent measurements, until both notches matched the designed hydraulic dimensions within acceptable tolerance.
- **Final Preparation:** After grinding, the notch edges were lightly deburred to restore sharpness (essential for accurate flow separation), and the plates were cleaned to remove any grinding residue or burrs.

Model:

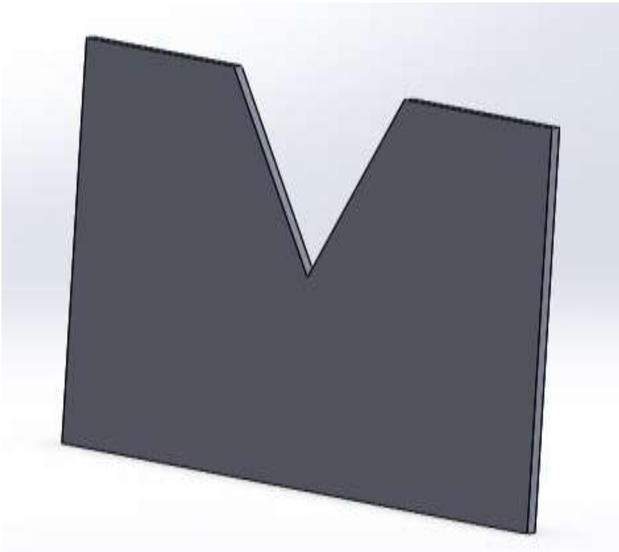


Figure 3

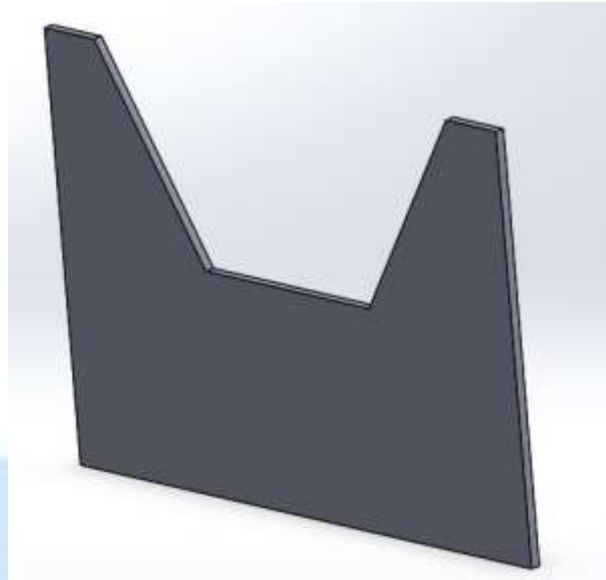


Figure 4

Figure 5

Figure 6

Experimental procedure

Procedure for Triangular Notch

- First, the hydraulic bench and all measuring instruments were checked properly to make sure that the apparatus was working correctly and there was no leakage in the system.
- The triangular notch plate was installed vertically at the outlet of the open channel. Proper alignment was ensured so that the notch remained exactly in the center.
- The water supply valve of the hydraulic bench was opened slowly, and water was allowed to flow through the channel.
- The flow was allowed to stabilize for some time until steady flow conditions were achieved and the water surface became stable.
- The crest level of the notch was noted as a reference point using the point gauge.
- The point gauge was carefully adjusted to measure the height of water above the notch crest.
- The head of water was recorded carefully at eye level to avoid parallax error.
- Water flowing through the notch was collected in the measuring tank.
- A stopwatch was used to measure the time required to collect a fixed volume of water.
- The collected volume of water was noted from the measuring tank scale.
- The experimental discharge was calculated using:

$$Q = \frac{V}{t}$$

- The flow rate was increased by adjusting the valve, and the same procedure was repeated for different head values.
- Three readings were taken for each flow condition, and the average values were recorded.

Procedure for Trapezoidal Notch

- After completing the experiment for the triangular notch, the water supply was turned off.
- The triangular notch plate was removed carefully from the setup.
- The trapezoidal notch plate was installed vertically in the same position with proper alignment.
- The hydraulic bench was started again, and water was allowed to flow through the channel.
- The flow was allowed to stabilize until steady flow conditions were achieved.
- The water head above the crest of the trapezoidal notch was measured using the point gauge.
- The measured head was recorded carefully.
- Water passing through the notch was collected in the measuring tank.
- The collection time was measured using a stopwatch.
- The collected volume of water was recorded.
- The experimental discharge was calculated using the volumetric method.

$$Q = \frac{V}{t}$$

- The water flow rate was increased gradually, and the same procedure was repeated for different heads.
- Multiple readings were taken, and average values were used for analysis.

Precautions

- Ensure the notch plate is installed vertically.
- Check that the notch edges are sharp and clean.
- Wait for steady flow before taking readings.
- Read the point gauge at eye level.
- Avoid leakage from the setup.
- Take multiple readings for better accuracy.
- This format usually matches how university project reports are written.

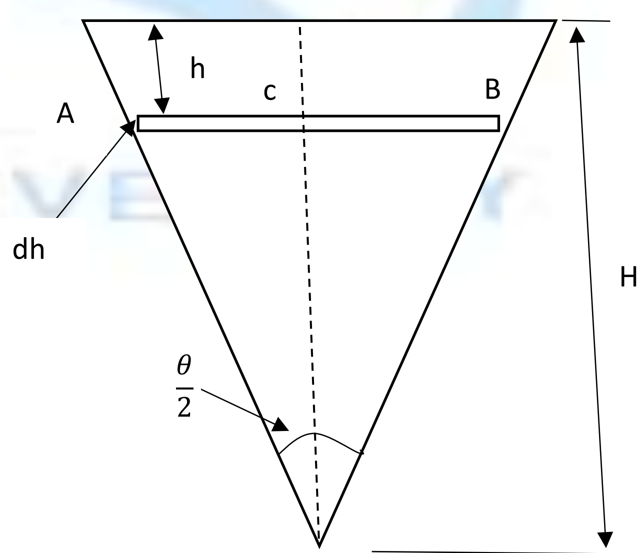
Nomenclature:

Column Heading	Units	Nom.	Type	Description
Notch Type	—	—	Measured	Vee Notch or Trapezoidal Notch
Height Datum	m	(h_o)	Measured	Datum height, which is the base of the notch. This is read from the vernier and used to calculate the height of water level above the notch. The height datum is measured in millimeters. Convert to meters for the calculations.

Water Level	m	(h)	Measured	This is read from the vernier. The water level is measured in millimeters. Convert to meters for the calculations.
Volume Collected	(m^3)	(V)	Measured	Taken from scale on hydraulics bench. The volume collected is measured in liters. Convert to cubic meters for the calculations (divide reading by 1000).
Time for Collection	s	(t)	Measured	Time taken to collect the known volume of water in the hydraulics bench. The time is measured in seconds.
Volume Flow Rate	(m^3/s)	(Q_t)	Calculated	(Q_t = Volume Collected / Time for Collection
Height above Notch	m	(H)	Calculated	($H=h-h_o$) = Height of Water Level − Height Datum
($H^{\frac{3}{2}}$) Trapezoidal Notch	—	—	Calculated	Used to describe relationship between flow rate and height for a Trapezoidal notch (rectangular portion only).
Trapezoidal Notch Discharge Coefficient	—	(C_d)	Calculated	$C_d = \dfrac{Q_a}{\sqrt{2g}\left[\frac{2bH^{3/2}}{3} + \frac{8}{15}\tan\frac{\theta}{2}H^{5/2}\right]}$
($H^{\frac{5}{2}}$) Vee Notch	—	—	Calculated	Used to describe relationship between flow rate and height for a Vee notch.
Vee Notch Discharge Coefficient	—	(C_d)	Calculated	$C_d = \dfrac{Q_a}{\frac{8}{15}C_d\sqrt{2g}\tan\frac{\theta}{2}H^{5/2}}$

Derivations and Calculations

Triangular Notch:



$$AB = 2AC$$

$$\tan\frac{\theta}{2} = \frac{AC}{H-h}$$

$$AC = (H-h)\tan\frac{\theta}{2}$$

$$AB = 2(H-h)\tan\frac{\theta}{2}$$

$$dA = AB \times dh$$

$$dA = 2(H-h)\tan\frac{\theta}{2}\ dh$$

$$v = \sqrt{2gh}$$

$$dQ = A \times v$$

$dQ = dA \times v$

$dQ = 2(H - h)\tan\frac{\theta}{2}\sqrt{2gh}\ dh$

$$dQ_a = C_d(dQ)$$

$$dQ_a = 2C_d(H - h)\tan\frac{\theta}{2}\sqrt{2gh}\ dh$$

$$Q_a = \int_0^H 2\,C_d(H - h)\tan\frac{\theta}{2}\sqrt{2gh}\ dh$$

$$Q_a = 2C_d\tan\frac{\theta}{2}\sqrt{2g}\int_0^H (H - h)\sqrt{h}\ dh$$

$$Q_a = 2C_d\tan\frac{\theta}{2}\sqrt{2g}\int_0^H (H\,h^{1/2} - h^{3/2})dh$$

$$Q_a = 2C_d\tan\frac{\theta}{2}\sqrt{2g}\left[H\frac{2h^{3/2}}{3} - \frac{2h^{5/2}}{5}\right]_0^H$$

$$Q_a = 2C_d\tan\frac{\theta}{2}\sqrt{2g}\left[\frac{2H^{5/2}}{3} - \frac{2H^{5/2}}{5}\right]$$

$$Q_a = 2C_d\tan\frac{\theta}{2}\sqrt{2g}\left[\frac{10 - 6}{15}H^{5/2}\right]$$

$$Q_a = 2C_d\tan\frac{\theta}{2}\sqrt{2g}\left[\frac{4}{15}H^{5/2}\right]$$

$$Q_a = \frac{8}{15}C_d\tan\frac{\theta}{2}\sqrt{2g}H^{5/2}$$

Coefficient of Discharge:

$$C_d = \frac{Q_a}{\frac{8}{15}\tan\frac{\theta}{2}\sqrt{2g}H^{5/2}}$$

Trapezoidal Notch:

$Q_a = C_d Q_{th}$

$Q_a = C_d A v$

AC = DB

$$\tan\frac{\theta}{2} = \frac{AC}{H-h}$$

$$AC = (H-h)\tan\frac{\theta}{2}$$

$$2AC = 2\tan\frac{\theta}{2}(H-h)$$

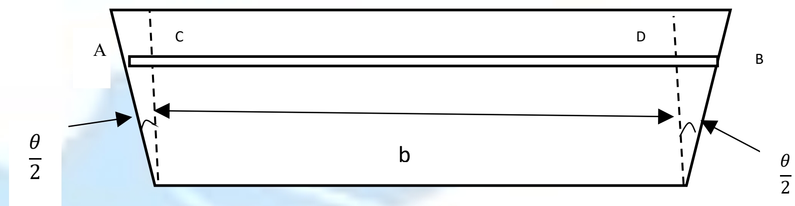
$$L = b + 2AC$$

$$L = b + 2\tan\frac{\theta}{2}(H-h)$$

$$A = L \times dh$$

$$A = [b + 2\tan\frac{\theta}{2}(H-h)]dh$$

$$v = \sqrt{2gh}$$



$$dQ_a = C_d A v$$

$$dQ_a = C_d[b + 2\tan\frac{\theta}{2}(H-h)]dh\sqrt{2gh}$$

$$dQ_a = C_d\sqrt{2g}[b + 2\tan\frac{\theta}{2}(H-h)]\sqrt{h}\,dh$$

$$Q_a = C_d\sqrt{2g}\int_0^H [b + 2\tan\frac{\theta}{2}(H-h)]\sqrt{h}\,dh$$

$$Q_a = C_d\sqrt{2g}\int_0^H [b + 2H\tan\frac{\theta}{2} - 2h\tan\frac{\theta}{2}]\sqrt{h}\,dh$$

$$Q_a = C_d\sqrt{2g}\int_0^H [b\,h^{1/2} + 2H\tan\frac{\theta}{2}h^{1/2} - 2\tan\frac{\theta}{2}h^{3/2}]dh$$

$$Q_a = C_d\sqrt{2g}\left[\frac{2bh^{3/2}}{3} + \frac{4H\tan\frac{\theta}{2}h^{3/2}}{3} - \frac{4\tan\frac{\theta}{2}h^{5/2}}{5}\right]_0^H$$

$$Q_a = C_d\sqrt{2g}\left[\frac{2bH^{3/2}}{3} + \frac{8}{15}\tan\frac{\theta}{2}H^{5/2}\right]$$

Coefficient of Discharge:

$$C_d = \frac{Q_a}{\sqrt{2g}\left[\frac{2bH^{3/2}}{3} + \frac{8}{15}\tan\frac{\theta}{2}H^{5/2}\right]}$$

Derivation is performed here and the sample calculations for coefficient of discharge is at the end of the report in the form of hand written calculations.

Results:

Notch Type	Datum Height h_o (mm)	h (mm)	Water Level $H=h-h_o$ (mm)	Volume Collected m^3	Time For Collection t (s)	Volume Flow Rate (Q_a)	Height Above Notch (H)	(cd)	$H^{\frac{3}{2}}$ (m)	$H^{\frac{5}{2}}$ (m)
Trapezoidal	10	31	21	0.005	9	0.00056	21	0.62	0.003	0.00006
	10	38	28	0.005	6	0.00083	28	0.59	0.005	0.00013
	10	40	30	0.005	3.48	0.0014	30	0.90	0.005	0.00016
	10	43	33	0.005	5	0.001	33	0.54	0.006	0.00020
	10	45	35	0.005	3.2	0.0016	35	0.76	0.007	0.00023
	10	26	16	0.005	16.4	0.00030	16	0.52	0.002	0.00003
	10	20.5	10.5	0.005	62	0.00008	10.5	0.27	0.001	0.00001
Triangular	10	47	37	0.005	38.4	0.00013	37	0.36	-	0.00026
	10	59	49	0.005	14.3	0.00035	49	0.48	-	0.00053
	10	66	56	0.005	9.76	0.00051	56	0.51	-	0.00074
	10	72	62	0.005	9	0.00056	62	0.43	-	0.00096
	10	75	65	0.005	8.5	0.00059	65	0.40	-	0.00108
	10	62	52	0.005	12	0.00042	52	0.50	-	0.00062
	10	44	34	0.005	85	0.00006	34	0.20	-	0.00021
	10	61	51	0.005	11.6	0.00043	51	0.54	-	0.00059

Sample Calculations:

Sample Calculation 1 Triangular Notch

$$H = 56\text{ mm} = 0.056\text{ m}$$

$$Q_a = \frac{0.005}{9.76} = 0.000512\ m^3/s$$

$$H^{5/2} = (0.056)^{2.5} = 0.000742$$

$$C_d = \frac{Q_a}{\frac{8}{15}\tan\left(\frac{60^\circ}{2}\right)\sqrt{2g}H^{5/2}}$$

$$C_d = \frac{0.000512}{\left(\frac{8}{15}\right)(0.5774)(4.429)(0.000742)}$$

$$\boxed{C_d = 0.51}$$

Sample Calculation 2 Triangular Notch

$$H = 49 \text{ mm} = 0.049 \text{ m}$$
$$Q_a = \frac{0.005}{14.3} = 0.000350\ m^3/s$$
$$H^{5/2} = (0.049)^{2.5} = 0.000531$$
$$C_d = \frac{0.000350}{\left(\frac{8}{15}\right)(0.5774)(4.429)(0.000531)}$$
$$\boxed{C_d = 0.48}$$

Sample Calculation 1 Trapezoidal Notch

$$H = 21 \text{ mm} = 0.021 \text{ m}$$
$$Q_a = \frac{0.005}{9} = 0.000556\ m^3/s$$
$$H^{3/2} = 0.00304$$
$$H^{5/2} = 0.0000639$$
$$C_d = \frac{Q_a}{\sqrt{2g}\left[\frac{2bH^{3/2}}{3} + \frac{8}{15}\tan\left(\frac{30^\circ}{2}\right)H^{5/2}\right]}$$
$$b = 0.24\ m, \tan\ 15^\circ = 0.2679$$
$$C_d = \frac{0.000556}{4.429\left[\frac{2(0.24)(0.00304)}{3} + \frac{8}{15}(0.2679)(0.0000639)\right]}$$
$$\boxed{C_d = 0.26}$$

Sample Calculation 2 Trapezoidal Notch

$$H = 28 \text{ mm} = 0.028 \text{ m}$$
$$Q_a = \frac{0.005}{6} = 0.000833\ m^3/s$$
$$H^{3/2} = 0.00469$$
$$H^{5/2} = 0.000131$$
$$C_d = \frac{0.000833}{4.429\left[\frac{2(0.24)(0.00469)}{3} + \frac{8}{15}(0.2679)(0.000131)\right]}$$
$$\boxed{C_d = 0.25}$$

Discussion of Results:

The experimental results obtained from both the triangular notch and trapezoidal notch showed that the discharge increased as the head of water increased. This behavior agrees with the theoretical principles of fluid mechanics because the velocity of flow through the notch depends directly on the water head. During the experiment, it was observed that even small changes in head produced noticeable changes in discharge values, especially in the triangular notch.

From the collected readings, the triangular notch showed better sensitivity at lower flow rates. Small variations in water level were easier to detect, which made the triangular notch more suitable for measuring low discharges accurately. However, as the flow rate increased, the discharge values also increased rapidly, and slight fluctuations in head affected the readings more noticeably.

On the other hand, the trapezoidal notch was found to handle larger flow rates more effectively. The discharge through the trapezoidal notch was comparatively higher because of its larger effective flow area. The flow pattern over the trapezoidal notch also appeared more stable during experimentation. Due to the combined rectangular and triangular shape, the notch reduced contraction effects and provided smoother discharge characteristics.

The experimental coefficients of discharge for both notches were lower than the ideal theoretical values. This difference occurred because theoretical equations assume ideal flow conditions, while actual experiments involve frictional losses, turbulence, viscosity effects, and minor leakage. In addition, practical limitations such as slight dimensional inaccuracies, human observation errors, and unsteady flow conditions also affected the final results.

During experimentation, it was also noticed that accurate head measurement was extremely important. A very small error in measuring the water level produced a significant change in the calculated discharge because discharge depends strongly on head. Therefore, steady flow conditions and careful use of the point gauge were necessary to obtain reliable readings.

Overall, the results showed that both notches are effective devices for open channel flow measurement, but their performance depends on the flow conditions. The triangular notch is more suitable for low discharge measurements where higher sensitivity is required, while the trapezoidal notch is more suitable for medium and higher flow rates where larger discharge capacity is needed. The experiment successfully demonstrated the practical relationship between head and discharge and verified the theoretical concepts of open channel flow measurement.

Sources of error

During experimentation, some deviations between theoretical and experimental values were observed. These differences may occur due to various practical limitations.

Instrumental Errors

- **Point Gauge Reading Error:** The point gauge is used to measure head above the notch crest. If the gauge is not calibrated properly, incorrect readings may occur.
- **Stopwatch Error:** Human reaction time while starting or stopping the stopwatch can affect time measurement.

- **Measuring Tank Calibration Error:** If the measuring tank dimensions are inaccurate, the calculated discharge will also be affected.

Human Errors

- **Parallax Error:** If readings are not taken at eye level, measurement errors may occur.
- **Recording Mistakes:** Incorrect recording of head, volume, or time values can affect final calculations.
- **Improper Observation:** Failure to wait for steady flow conditions can produce inaccurate results.

Experimental Setup Errors

- **Improper Alignment of Notch Plate:** If the notch is not installed vertically, the flow pattern changes.
- **Leakage in Setup:** Water leakage reduces the actual discharge through the notch.
- **Vibrations:** Mechanical vibrations can disturb the free water surface and affect head readings.

Flow Related Errors

- **Unsteady Flow:** Fluctuations in water supply cause variation in head.
- **Turbulence:** Excess turbulence near the notch changes flow characteristics.
- **Air Entrapment:** Air bubbles may disturb smooth flow.

Fabrication Errors

- **Rough Edges:** If the notch edges are not sharp, flow separation changes.
- **Incorrect Dimensions:** Small dimensional inaccuracies affect discharge calculations.
- **Surface Irregularities:** Uneven surfaces increase flow resistance.

Industrial and practical applications

Notches are widely used in hydraulic engineering because of their simplicity, reliability, and low installation cost.

Applications of Triangular Notch

- Laboratory Flow Measurement: Triangular notches are widely used in fluid mechanics laboratories because they provide accurate readings for small flow rates.
- Research and Development: Researchers use triangular notches when studying low-flow hydraulic behavior.
- Small Irrigation Systems: Used to monitor water supplied to farms and agricultural land.
- Leak Detection Systems: Used where small discharge measurement is required.
- Chemical Processing Plants: Used for controlled low-flow liquid measurements.
- Pharmaceutical Industries: Used where precise fluid dosing is necessary.
- Cooling Water Monitoring: Used in small cooling systems.

Applications of Trapezoidal Notch

- Irrigation Canals: Used to measure water supplied to agricultural channels.
- Wastewater Treatment Plants: Used to monitor sewage and treatment flow.

- Hydroelectric Power Plants: Used for measuring water entering turbine systems.
- Municipal Water Supply Systems: Used to monitor water distribution.
- Industrial Drainage Systems: Used for discharge monitoring.
- Mining Industry: Used for slurry and water discharge measurement.
- Flood Monitoring Channels: Used to measure water levels and discharge.
- Reservoir Management: Used for controlled release measurements.

Applications in Mechanical Engineering

Notch-based flow measurement is useful in:

- Cooling circuits
- Heat exchanger systems
- Pump testing setups
- Hydraulic machinery
- Experimental fluid mechanics labs
- Prototype hydraulic systems

Applications in Civil Engineering

Notches are used in:

- Canal design
- Spillway analysis
- Storm water systems
- Dam monitoring
- Drainage networks

Applications in Environmental Engineering

Notches help in:

- Pollution monitoring
- Effluent measurement
- Water conservation systems
- River flow monitoring
- Wetland management

RECOMMENDATIONS

- Use precisely machined notch edges.
- Ensure proper calibration before testing.
- Perform multiple trials.
- Maintain steady flow conditions.
- Use digital measurement instruments for higher accuracy.

References:

- FM lab Report 4.pdf
- https://www.codecogs.com/library/engineering/fluid_mechanics/notches/trapezoidal-notch.php
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CONCLUSION

This project successfully demonstrated the practical use of **Triangular Notch** and **Trapezoidal Notch** for discharge measurement in open channel flow. It was observed that notch geometry directly affects sensitivity, operating range, and measurement accuracy. The triangular notch showed superior performance for lower flow conditions, whereas the trapezoidal notch was found to be more suitable for moderate and higher discharges. The experimental observations supported the theoretical principles of fluid mechanics and provided practical insight into hydraulic flow measurement systems.